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Guidelines

Aspiration pneumonia guidelines – Société de Pathologie Infectieuse de Langue Française 2025

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1. Introduction

Aspiration pneumonia (AP) is a bacterial pneumonia characterized by its physiopathological mechanism. French guidelines on this nosological entity have never been published. The need to combat antibiotic resistance, the evolution of the care pathway, and the medical progress made over the last years – both in terms of imaging tests and microbiological diagnosis – shed new light on pneumonia; thus, justifying the present guidelines. Community-acquired pneumonia (CAP) include a significant proportion of viral pneumonia, as well as a large number of bacterial pneumonia caused by micro-aspiration of ENT flora, and bacterial pneumonia acquired through inter-human transmission, of environmental origin, or secondary to macro-aspiration. We are currently witnessing an excessive prescription of antibiotics due to aspiration pneumonia diagnoses that are not based on robust diagnostic criteria and that are in fact excessively treated uncomplicated pulmonary aspiration cases. Rapid molecular diagnostic tests now allow for better identification of viral pneumonia, while low-dose chest CT-scans allow for ruling out dyspnea not associated with infectious pneumonia. These new tools should lead to a more precise diagnosis of bacterial pneumonia and to limiting unjustified antibiotic prescriptions.

Post-stroke pneumonia, early post-operative pneumonia, and ventilator-associated pneumonia are not addressed in the present guidelines, as are pneumonia associated with severity criteria requiring intensive care unit or continuing care unit admission.

The present guidelines are based on a literature analysis, followed by drafting of proposals by the expert group which were collectively validated by the writing group and an external reviewing group.

As there is no robust scientific studies, and especially no randomized controlled trials, the present guidelines are mainly based on experts' opinion in light of the few retrospective studies performed in countries other than France. The recommendations are classified according to various grades (Grade A, B, or C) depending on the level of scientific evidence found in the literature and according to three levels depending on the level of experts' support (**Table S1**).

2. Epidemiology chapter

Question 1: Definitions and microbiology of AP.

Definitions:

– **Chemical pneumonitis** is defined by the aspiration of gastric contents, that may lead to early inflammatory reaction with no bacterial origin. Unlike aspiration pneumonia, there is no incubation period [1].

– **Mechanical bowel obstruction** is defined by the aspiration of fluids or materials obstructing the airways or triggering a reflex called "airway closure", with no associated inflammatory reaction of the parenchyma.

– **Aspiration pneumonia (AP)** is defined by an active infection caused by the significant inoculation in the lungs of aerobic, anaerobic, or mixed bacterial populations.

Aspiration pneumonia is defined by its physiopathological mechanism, as opposed to acute community-acquired pneumonia and healthcare-associated pneumonia which are defined according to the place of acquisition (Fig. 1). AP corresponds to pulmonary involvement caused by oropharyngeal or gastric fluids entering the lower respiratory tract [2]. Epidemiological data on AP is scarce and of low scientific evidence because of the absence of robust diagnostic criteria. AP is estimated to account for 5 % to 15 % of community-acquired pneumonia but data is missing on hospital-acquired pneumonia [1]. Aspiration of small amounts of oropharyngeal secretions is normal in healthy people during their sleep [3]. Thus, causal mechanisms may be micro-aspiration but also macro-aspiration. The clinical presentation thus depends on the following variables: bacterial virulence, repeat aspirations, and place of acquisition.

The first studies on AP microbiology that identified a high proportion (61–92 %) of anaerobes were performed in the 1970s [4,5]. At that time, the main bacteria identified were *Fusobacterium* sp., followed by species of the *Bacteroides*, *Peptococcus*, and *Peptostreptococcus* genera [6,7]. The proportion of anaerobic bacteria reported in the literature then decreased (1.6 %–16 %), some studies even reported the absence of those bacteria [8,9] despite adequate sampling techniques [10–13]. In an international multicentric study of patients hospitalized for CAP,

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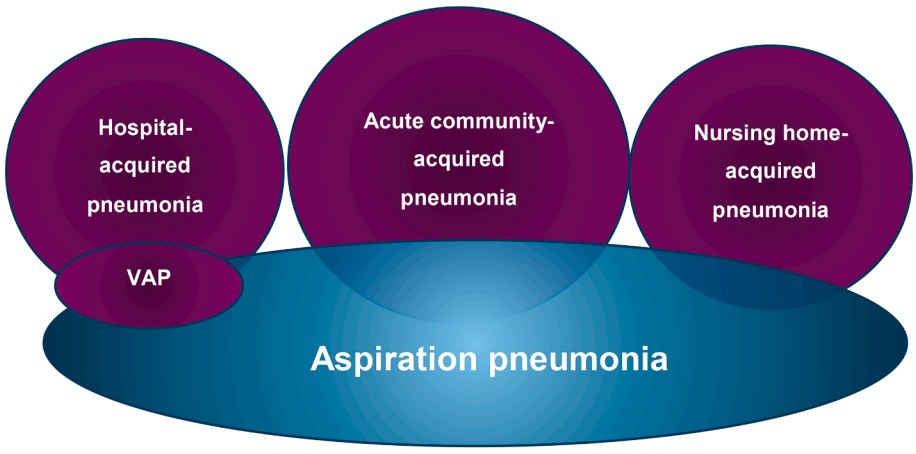


Fig. 1. Positioning of aspiration pneumonia (AP) among the different types of pneumonia. VAP: Ventilation-acquired pneumonia.

those with AP or with CAP and aspiration risk factors had an anaerobic flora similar to that of patients with no aspiration risk factors [11]. Currently, the most common bacteria identified in AP are *S. pneumoniae, H. influenzae, S. aureus*, some Enterobacterales, and *P. aeruginosa* [13].

The current epidemiology of AP seems to reflect bacterial colonization of the upper respiratory tract. This evolution could be linked to changes in sociodemographic characteristics and microbiological documentation methods. It is worth noting that no bacteria are isolated in cultures in approximately 30 % to 40 % of cases [4,6,10,11,14–21].

Question 2: What are the risk factors for AP?

Answer 2: Risk factors for AP are grouped into four main mechanisms: swallowing disorders, attention disorders, factors associated with gastric reflux, and anomaly of pharyngeal reflexes (cough and gag reflexes) (Table S2).

Arguments:

Aspiration of small amounts of oropharyngeal secretions is physiological, especially during sleep. Often silent, these micro-aspirations can lead to pneumonia especially in the elderly and following a stroke [2,22]. Poor oral hygiene (dry mouth, periodontal infection) favors the onset of infections by increasing the bacterial inoculum. Macro-aspirations are more often symptomatic, and risk factors can be grouped into four main mechanisms: attention disorders, swallowing disorders, factors associated with gastric reflux, and anomaly of pharyngeal reflexes (cough and gag reflexes) (**Table S2**) [2,22,23]. Risk factors are often intertwined and pile up, especially in the elderly. Undernourishment and sarcopenia, loss of functional independence and 'frailty' are additional risk factors for AP [2,24]. Aspiration risk factors identification is required to establish the AP diagnosis.

3. Diagnosis chapter

Question 3: What are the criteria for confirming AP diagnosis?

Recommendation 3:

1) AP is defined by the combination of (Table 1***):***

a) ***a positive diagnosis of pneumonia:*** **combination of clinical signs of acute lower respiratory tract infection and new radiological infiltrate(s);**

b) ***presumptive etiological diagnosis of aspiration:*** **evidence of aspiration or swallowing disorders and/or dorsal clinico-radiological involvement with risk factors for aspiration (attention disorders, swallowing disorders, anomaly of cough and gag reflexes, upper airways involvement, or alteration of oral hygiene).**

2) Acute respiratory signs just after pulmonary aspiration (chemical pneumonitis) do not correspond to the definition of AP (Grade C-1).

Arguments:

Acute pneumonia is defined by the combination of acute infection symptoms of the lung parenchyma (fever or hypothermia, chills, sweating, chest pain, new cough, new dyspnea) and by the presence of new radiological infiltrates [25].

Aspiration is defined by the aspiration of gastric or oropharyngeal contents into the larynx and the lower respiratory tract [1].

Aspiration pneumonia (AP) corresponds to the bacterial proliferation and invasion of the lung parenchyma following the usually silent [26] aspiration of secretions colonized by pathogenic bacteria [27].

Table 1
Diagnostic criteria for aspiration pneumonia.

Diagnosis of AP: (1) + (2)
(1) **Positive diagnosis of pneumonia**
+
- Clinical signs of acute lower respiratory tract infection
- New radiological infiltrate(s)
(2) **Presumptive etiological diagnosis of aspiration**
or
(attention disorders, swallowing disorders, anomaly of cough and gag reflexes, upper airways involvement, or alteration of oral hygiene)
- Swallowing disorders or aspiration
- Clinical and radiological dorsal involvement with risk factor(s) for aspiration

AP: aspiration pneumonia.

AP should be distinguished from aseptic chemical pneumonitis, which occurs just after aspiration of gastric contents (Mendelson's syndrome) [26] or of a foreign body (Foreign body aspiration) [28]. Mendelson's syndrome and foreign body aspiration only rarely lead to infectious pneumonia [29].

There is currently no gold standard criteria for the diagnosis of aspiration. It is most often a presumptive diagnosis based on anamnestic elements (search for risk factors, **Table S2**), and to a lower extent on clinical (identification of swallowing disorders) and radiological criteria (see below) [30].

Question 4: Is radiological confirmation required for the diagnosis of AP?

Recommendation 4:

An imaging test is recommended to confirm the diagnosis of AP (Grade B-1).

Arguments:

The low specificity of the clinical examination does not allow for establishing the diagnosis of AP without any radiological confirmation [25,30]. In very elderly patients, the diagnostic performance of the clinical examination for the diagnosis of acute pneumonia is low [31,32] especially because of atypical symptoms. The authors of a study of 2,334 residents of 36 nursing homes in Missouri, United-States, with a clinical diagnosis of pneumonia, reported that 45 % of residents presented with radiological signs of pneumonia (probable diagnosis in 12 % of cases, and confirmed diagnosis in 35 %) [33].

In addition, antibiotic prescription to patients presenting with clinical signs compatible with a diagnosis of AP without any radiological confirmation is responsible for an excessive prescription of antibiotics, especially during winter.

The authors thus recommend confirming the diagnosis with systematic lung imaging in case of AP suspicion, consistent with the current recommendations for lower respiratory tract infections [34].

Question 5: Which imaging tests must be performed in patients with AP suspicion?

Recommendation 5:

It is recommended to primarily perform a chest CT-scan without injection to establish a positive radiographical diagnosis of AP. Otherwise, lung ultrasound or chest radiography can be performed in case of difficult access to a CT-scan (Grade B-1).

Arguments:

The diagnosis of AP is based on clinical criteria, the presence of risk factors, and evocative radiological results, i.e., infiltrates that depend on severity (dorsal involvement). Dorsal involvement is defined by basal segments involvement of the lower lobes while standing and seating, especially of the right lower lobe; by posterior segments involvement of the right upper lobe and apical segment involvement of the right lower lobe in supine position [23,35,36].

Similar to other types of pneumonia, chest radiography results can sometimes be difficult to interpret – especially in elderly people and in bedridden patients – and images may appear with a delay after the onset [23,37]. Up to 28 % of chest CT scan-confirmed AP cases have normal chest radiography results [34,38]. In elderly patients with acute pneumonia suspicion, there is also major variability in the interpretation of the chest radiography results depending of the clinician [39]. Chest CT-scan without injection is currently the most performing examination for the radiological diagnosis of AP [40]. Lung ultrasound has not been extensively assessed in AP. Only one Japanese monocentric study compared results from chest radiography, CT-scan, and ultrasound in 34 patients, and reported strong correlation between the ultrasound and the CT-scan results, but not with the radiological results [41]. Ultrasound performance for the diagnosis of acute pneumonia is close to that of the CT-scan when performed by an experienced operator [42–44], enabling diagnostic confirmation at the patient's bed.

Considering the atypical symptoms in pneumonia of the elderly and the low negative predictive value of conventional radiography, performing additional imaging tests does not contradict acute community-acquired pneumonia guidelines. Lung ultrasound has not been sufficiently assessed in this specific indication.

Question 6: What is the role of biological markers in the diagnosis of AP?

Recommendation 6:

The use of biological markers is not recommended to confirm or to rule out the diagnosis of AP (Grade C-2).

Arguments:

The use of biological markers to distinguish acute pneumonia non-associated with aspiration, AP, and chemical pneumonitis has not been extensively assessed. Some studies reported a significant increase in procalcitonin, CRP, and leucocyte levels in patients with AP compared with patients with aspiration but without bacterial pneumonia; other studies did not report any significant differences [45–48]. Increase in procalcitonin levels has been reported in non-bacterial chemical pneumonitis. Compared with other types of pneumonia, AP is supposed to be associated with a lower CRP level [10,45,46,48–53].

4. Microbiology chapter

Question 7: Which microbiological tests are recommended for diagnostic purposes?

Recommendation 7:

- **No systematic additional microbiological test is recommended (Grade C-1).**
- **Sputum cytology can be performed before initiating the antibiotic therapy active against *Pseudomonas aeruginosa* (Grade C-1).**
- **Urine antigen tests (*Streptococcus pneumoniae* and *Legionella*) are not recommended in the diagnosis of AP (Grade C-2).**

Arguments:

Direct examination and culture of respiratory isolates, despite their low specificity and sensitivity [54], ensure the epidemiological monitoring of AP [13] and, in some cases, allow for adjusting the antibiotic therapy if a resistant or atypical pathogen is identified [9,55,56]. However, given the lack of evidence of mortality improvement at the individual level and because of the difficulty in performing good quality examination [57,58], obtaining the results of these tests must not delay antibiotic therapy initiation. Invasive sampling must be avoided. Finally, because of the heterogeneity of the population concerned, the decision to collect samples must be assessed by the physician in charge and depends on the clinical state of the patient and on the treatment plan.

It should be reminded that, during an epidemic, the diagnosis of viral pneumonia should be considered, and a virological diagnostic test should be performed.

5. Treatment chapter: antibiotic therapy

Question 8: What is the first-line antibiotic therapy?

Recommendation 8:

- **First-line treatment:**

- **First choice: amoxicillin-clavulanic acid, oral or intravenous (IV) (Grade B-1)**
- **Second choice (when the first choice cannot be used, especially in patients with an allergy to penicillin without any contraindication to third-generation cephalosporins (3GC) and/or when the oral or IV route cannot be used): subcutaneous (SC) ceftriaxone (Grade B-1)**
- **Severe allergy to beta-lactams: co-trimoxazole (Grade C-2)**

Arguments:

The spectrum of the molecule must cover Enterobacterales, methicillin-susceptible *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Haemophilus*. As anaerobic bacteria are involved in less than 5 % of AP cases, their specific coverage is not required except for the specific cases identified in the Infectious Diseases Society of America (IDSA) guidelines (obvious dental/periodontal diseases, lung abscess, or empyema complications) [11,59]. A difficult balance must be found between the sufficient coverage of the polymicrobial nature of AP and the risk of an excessive use of broad-spectrum antibiotics that favors the emergence of antibiotic resistance, *Clostridioides difficile* colitis [9,18,60], and an excessive mortality rate due to the excessive use of broad-spectrum antibiotics [61,62]. Several studies compared amoxicillin or amoxicillin-clavulanic acid with 3GCs in the treatment of acute community-acquired pneumonia, including a number of AP cases, but the authors did not conclude to the superiority of cephalosporins [63,64]. Several studies performed in Germany and Japan compared treatment plans based on ampicillin-sulbactam with ceftriaxone or with moxifloxacin, and a combination of clindamycin and a cephalosporin for AP and lung abscesses [8,13,65,66]. Although clinical trial data supporting the effectiveness of 3GCs in AP is available, the use of 3GCs should not be favored in order to minimize adverse ecological events associated with their use. Thus, the combination of ampicillin-sulbactam or amoxicillin-clavulanic acid (more commonly used in France) is the reference antibiotic therapy for AP, consistent with the various international guidelines (Japanese, British Thoracic Society, and Infectious Diseases Society of America) [59,67,68]. Expert opinion from Japan and from the British Thoracic Society (BTS) is even in favor of the use of ampicillin or amoxicillin without any beta-lactamase inhibitor as the first-line treatment for ambulatory forms [67,68]. A French study of ambulatory prescription databases reported that, among more than 50,000 patients aged over 75 years treated for bacterial pneumonia in community settings, 33.5 % received amoxicillin-clavulanic acid and 21.3 % received amoxicillin [69].

Ceftriaxone is the second-line treatment, not combined with metronidazole (no need to cover anaerobic bacteria of the *Bacteroides* genus), in the following two indications:

1) Patients who cannot be treated with amoxicillin-clavulanic acid because of a penicillin allergy, without any contraindication to the use of 3GCs.

2) Patients who cannot receive medications through the oral or intravenous routes (dementia, agitation, severe swallowing disorders, etc.). The administration of subcutaneous ceftriaxone as one injection per day ensures treatment compliance while limiting iatrogenic effects linked to vascular devices [70–72].

For patients with contraindications to penicillins and 3GCs, co-trimoxazole is suggested as an alternative. Its spectrum meets the dual objective of covering the most frequently isolated pathogens (especially Enterobacterales) – with better susceptibility rates reported in France than those of other alternative antibiotics (doxycycline, clindamycin, pristinamycin) – and of being classified among antibiotics with a "preferential use" in community settings as well as in hospital settings [73]. Data from the French National Reference Center for Pneumococci shows 92.8 % susceptibility of pneumococci to co-trimoxazole in 2021 versus 78.9 % for tetracyclines [74].

- **For AP with severity criteria (Table S3):**
  - **It is recommended to consider risk factors for infection that could involve *Pseudomonas aeruginosa* and methicillin-resistant *Staphylococcus aureus* (MRSA) (Grade B-1).**
  - **Piperacillin-tazobactam is recommended as the empirical treatment for patients with at least one of the following risk factors: documented history of recent respiratory tract colonization or infection (<1 year) due to *Pseudomonas*, recent parenteral antibiotic therapy (<3 months), severe chronic obstructive pulmonary disease (COPD), bronchiectasis, tracheostomy (Grade B-1).**
  - **Adding linezolid is recommended in the empirical treatment of patients with previous documented colonization with MRSA (Grade B-1).**
  - **In the absence of risk factors for *Pseudomonas aeruginosa* or MRSA infection, the recommended treatment is similar to that recommended for forms without any severity criteria (Grade B-1).**

Arguments:

The study by Martin-Coral [11] reported an increased rate of *Pseudomonas* infections in patients with AP and severity criteria. When severity criteria are observed, the empirical antibiotic therapy spectrum needs to be broadened when risk factors for *Pseudomonas* and MRSA infections uncovered by the first-line amoxicillin-clavulanic acid antibiotic therapy are present. Validated factors associated with the occurrence of MRSA and *Pseudomonas* infections are similar to those of patients with CAP and severity criteria. Validated risk factors associated with *Pseudomonas* pneumonia are history of recent documented *Pseudomonas* respiratory tract colonization or infection (<1 year), recent parenteral antibiotic therapy (<3 months), severe COPD, bronchiectasis, and tracheostomy. Validated risk factors associated with MRSA pneumonia is previous colonization with MRSA, with no specification of time [75–77]. The community or hospital origin of the pneumonia has no direct impact on the antibiotic therapy choice [78,79].

- **For pneumonia with severity criteria (Table S3) and indeterminate mechanism:**
  - **It is recommended to follow the 2025 SPILF guidelines on severe CAP.**

Arguments:

The diagnosis of AP is based on a number of factors, including a suspicion of aspiration mechanism. This suspicion is sometimes difficult to validate in the absence of imaging tests available in an urgent setting or when the patient's history cannot be collected or is not contributory. Thus, the presence of severity criteria and of an indeterminate mechanism should lead physicians to follow the guidelines on severe CAP. There is no literature data specific to AP that justifies coverage of intracellular bacteria when confronted to an aspiration mechanism despite the presence of severity criteria [11,52].

- **Following pulmonary aspiration:**
  - **It is recommended not to start preemptive antibiotic therapy but to proceed to clinical monitoring and to start the antibiotic therapy only in the presence of AP criteria (Grade C-1).**

Arguments:

AP occurrence following pulmonary aspiration is rare. Cough and dyspnea usually resolve within hours. Aspiration models following cardiac arrest or neurological coma did not demonstrate the benefit of a prophylactic or preemptive treatment for AP on mortality or hospital length, even though such treatment reduces the number of ventilation-acquired pneumonia [80,81]. In order to avoid unnecessary exposure to antibiotic therapies, the expert group recommends a wait-and-see attitude, only treating AP with diagnostic criteria (please refer to Question 3). When no acute respiratory distress syndrome is observed, it

is reasonable to wait 48 to 72 h before initiating the antibiotic therapy.

- **For known MRSA or *Pseudomonas aeruginosa* bronchial colonization:**
- **The choice of empirical antibiotic therapy requires specialist advice.**
- **When no positive clinical course is observed 72 h after antibiotic therapy initiation:**
- **A treatment switch to piperacillin-tazobactam is suggested (Grade B-1).**

Arguments:

Assessment should be conducted at 72 h and not at 48 h, because of the slower progression of symptoms in elderly patients, in order to avoid classifying as "treatment failure" a slow clinical course that is in the end positive. The 72-hour assessment is useful to check whether the stability criteria defined on Day 3 for acute community-acquired pneumonia are met. Antibiotic therapy failure can only be considered when such criteria are not observed [82]. In case of suspected microbiological failure, differential diagnoses of viral pneumonia and pneumonia caused by intracellular bacteria should first be ruled out as these diagnoses do not fall within the scope of AP. A bacterial origin not covered by the initially prescribed treatment should then be considered as well as non-infectious differential diagnoses. The various microbiological studies performed reported that *P. aeruginosa* or Group 3 Enterobacteriaceae not covered by amoxicillin-clavulanic acid are isolated in less than 10 % of AP cases with no severity criteria [11,13]. Based on clinical experience and on the expert group's opinion, we suggest to switch amoxicillin-clavulanic acid to piperacillin-tazobactam following respiratory sample collection for quantitative microbiological analysis purposes, and then to adjust the antibiotic therapy to microbiological results.

Criteria for a positive clinical course:

The positive clinical course is evaluated at 72 h by the presence of all of the following clinical stability criteria:

- Temperature $\leq$ 37.8 °C
- Systolic blood pressure $\geq$ 90 mm Hg
- Heart rate $\leq$ 100/min
- Respiratory rate $\leq$ 24/min
- SpO2 $\geq$ 90 % or PaO2 $\geq$ 60 mm Hg in ambient air

Persistence of cough is not a criterion of non-improvement.

- **For relapses and recurrences of AP with previous positive clinical course:**
- **It is suggested to newly initiate the same antibiotic regimen (Grade C-1).**

Arguments:

Relapses or recurrences within 30 days of the initial episode are expected in 10 to 30 % of cases. This new episode can occur during the same hospital stay period or after having been discharged home or in a rehabilitation facility; AP is in that case classified as hospital-acquired or healthcare-associated pneumonia. Fifteen years ago, hospital-acquired and healthcare-associated pneumonia were considered at risk of being caused by multidrug-resistant bacteria, but numerous data have called these assumptions into question [78,83–86]. There is no French data available to define the epidemiology of the resistance of bacteria responsible for hospital-acquired or healthcare-associated relapses or recurrences. Based on clinical experience and on the expert group's opinion, it is suggested to newly initiate the same antibiotic regimen prescribed for the first episode following respiratory sampling for microbiological purposes, and then to adjust the antibiotic therapy to microbiological results.

- **The choice of the molecule and dosing regimen used for the first-line treatment can be adapted, based on the infectious disease specialist's advice and taking into consideration the patient's individual factors such as any recent exposure to antibiotics and recent microbiological results.**
- **Criteria for hospitalization are those of the CAP guidelines.**
- **Dosing regimens are proposed in Table 2.**

Question 9: Should metronidazole be used in the treatment of AP?

Recommendation 9:

Combining metronidazole with the reference antibiotic therapy for AP is not recommended (Grade C-1).

Arguments:

Recent studies reported a major proportion of *Streptococcus pneumoniae* and *Haemophilus influenzae* in community-acquired forms of AP. The proportion of Enterobacterales (up to 55 % in some studies), *Pseudomonas aeruginosa*, and *Staphylococcus aureus* is higher in hospital-acquired AP. Anaerobic bacteria have long been considered as the main pathogens responsible for AP, but their role is now being discussed, and their prevalence seems to be lower than initially estimated. Documented anaerobic infections account for 2 % to 4 % of AP cases today; systematic coverage is therefore not justified. It should however be reminded that all of the molecules recommended in the treatment of AP are active against most anaerobic bacteria of the oral sphere. In a systematic review, Yoshimatsu *et al.* did not report any microbiological data justifying the use of a systematic anti-anaerobic treatment nor any benefit of such treatment in AP [87]. Bai *et al.* performed a large retrospective study and reported the absence of benefit on mortality of adding metronidazole or clindamycin, but they observed a higher incidence of *Clostridioides difficile* colitis [88]. Lastly, recent data showed that excess mortality is favored by the use of antibiotic therapies that are too broad-spectrum or cover too many anaerobic bacteria [61,62].

However, consistent with the IDSA and BTS guidelines, we highlight the importance of anti-anaerobic coverage in patients with AP and obvious dental/periodontal disease or in those with a suspicion of complication such as lung abscess or empyema; metronidazole should in that case be added to antibiotics with low activity against anaerobes such as co-trimoxazole and levofloxacin. The other antibiotics used in the treatment of AP (amoxicillin-clavulanic acid, piperacillin-tazobactam, and ceftriaxone) are sufficiently active against the potentially involved anaerobes and metronidazole should not be added when such antibiotics are prescribed. Bacteria of the *Bacteroides* genus, potentially resistant to these antibiotics (especially to ceftriaxone), are only very rarely isolated [11,59,67].

Question 10: Which administration routes should be used?

Recommendation 10:

- **It is recommended to use the oral route as a priority (Grade B-1).**
- **Subcutaneous ceftriaxone is an alternative**
  - o **When the oral route cannot be used**
  - o **When the IV route cannot be used**
  - o **As a switch from the IV route after stabilization and when an oral switch is impossible**

(Grade B-1)

- **The IV route should only be used for severe pneumonia cases**

Arguments:

Oral treatment of acute community-acquired pneumonia is well codified and the IV route is only used when the oral route cannot be used or when urgent administration of antibiotics is required because of severe clinical signs and symptoms [59]. No study has specifically assessed the benefit of the initial IV route for the treatment of AP. The post hoc analysis of the PTC randomized controlled trial [89] showed that, among hospitalized patients with CAP, there was no significant

Table 2
Summary of AP treatments.

	Molecule	According to GFR	Dosing regimen	Administration route
First line	Amoxicillin 1 g-clavulanic acid 125 mg	>30 ml/min	1 g of amoxicillin every 8 hrs	Oral or IV (if severity criteria)
		[10–30] ml/min	1 g as a loading dose, then 0.5 g every 8 hrs	
		<10 ml/min	1 g as a loading dose, then 0.5 g every 24 hrs	
Second line (if allergy to penicillin or oral/IV routes impossible)	ceftriaxone 1 g	No adaptation	1 g every 24 hrs	SC
Third line (if true and severe allergy to beta-lactams)	Sulfamethoxazole 800 mg/ trimethoprim 160 mg	>30 ml/min	800 mg of sulfamethoxazole every 8 hrs	Oral or IV
		[15–30] ml/min	800 mg every 24 hrs	
		<15 ml/min	800 mg every 48 hrs	
In case of treatment failure at 72 hrs	Piperacillin 4 g-tazobactam 500 mg	>20 ml/min	4 g of piperacillin every 6 to 8 hrs	IV
		<20 ml/min	4 g every 12 hrs	

For obese patients, the daily dosage must be adapted using the online tool ABXBMI [153].

difference in terms of effectiveness between the initial IV treatment and the oral-only treatment.

Published data on the SC use of ceftriaxone showed 100 % bioavailability, with PK/PD target achievement similar to the IV route. SC ceftriaxone can be used in light of available data on its benefit/risk ratio, which is deemed favorable in the above-mentioned indications. Prescribers must inform patients and/or their family of this off-label use before administering it, collect consent, explain the prescription in the medical file, and implement close monitoring of antibiotic tolerability, especially at the injection site [70,90].

Question 11: What is the total treatment duration?

Recommendation 11:

The recommended treatment duration if 5 days in case of positive clinical course (Grade C-1).

- **The positive clinical course is evaluated at 72 hours by the presence of all the following clinical stability criteria:**
  - **Temperature ≤37.8°C**
  - **Systolic blood pressure ≥ 90 mm Hg,**
  - **Heart rate ≤ 100/min,**
  - **Respiratory rate ≤ 24/min,**
  - **SpO2 ≥ 90 % or PaO2 ≥ 60 mm Hg in ambient air;**
- **Persistence of cough is not a criterion of non-improvement.**

Arguments: No randomized study has specifically studied treatment duration in AP. In their randomised controlled study [91], Dinh *et al.* compared three-day treatment duration with eight-day duration, but the results cannot be extrapolated to our situation despite the high mean age of the population studied because AP was an exclusion criterion in Dinh *et al.*'s study. However, several *meta*-analyses of studies with AP cases compared the effectiveness of antibiotic treatments longer than 7 days with 3- to 7-day treatments without identifying any differences in terms of effectiveness [92–95]. The expert group thus advise in favor of a 5-day treatment duration in case of a positive clinical course. Treatment duration should not exceed 7 days when local complications are not objectively observed (abscess, empyema).

6. Treatment chapter: Medications other than antibiotic therapies

Question 12: Other than antibiotics, which drugs should be used to treat AP?

Recommendation 12:

- **No specific drug treatment other than antibiotics is recommended considering the current state of knowledge.**
- **Corticosteroids are not recommended in the combined treatment of AP (Grade C-1).**

Arguments:

Corticosteroids were one of the first treatments described in the treatment of chemical pneumonitis in the 1950 s [1]. However, most data available in the literature date back from before the 1980 s, are of low scientific evidence, and include conflicting data on indirect benefits (potential improvement of imaging results or inflammatory parameters) [96] and adverse events often more important than the expected benefit: prolonged hospital stay, glycemic imbalance, confusion or hydro-retention or even, in some studies, a higher risk of Gram-negative bacilli infections [1,97,98]. Several studies with high scientific evidence demonstrated the benefit of corticosteroid therapy in severe acute community-acquired pneumonia, acute respiratory distress syndrome, or septic shock. Aspiration pneumonia was excluded from these studies; making it impossible to recommend corticoids in this indication [81,99–103].

- **Aerosols are not recommended in the combined treatment of AP. They can be prescribed to patients with evidence of bronchospasm, ideally with bronchodilator aerosols (Beta$_2$-adrenergic agonists) (grade C-1).**
- **Off-label scopolamine is not recommended, except for the treatment of agonic rales due to salivary congestion at the end of life (grade C-1).**

7. Prevention chapter: drug therapies

Question 13: Which drug adaptations can be useful to prevent AP?

Recommendation 13:

The preferential use of ACE inhibitors over other antihypertensive agents is recommended in patients who have had a stroke, especially in those of Asian origin (grade C-1).

Arguments:

Following a stroke, the use of ACE inhibitors over other antihypertensive agents could reduce the risk of AP especially in patients of Asian origin. The hypothesized mechanism of action of ACE inhibitors is an increase in the level of substance P, which promotes the cough reflex [2]. A randomized study [104] assessing perindopril versus placebo showed a downward trend in the occurrence of pneumonia, without reaching significance in the general population. The subgroup analysis showed a significant reduction in patients of Asian origin. A *meta*-analysis [105] showed that, compared with other antihypertensive drugs, ACE inhibitors lowered the risk of post-stroke pneumonia (Odd Ratio 0.6), especially in Asia (OR 0.42) and more particularly in Japan

(OR 0.38) [105]. Other cohort studies did not observe such effect [106,107]. European guidelines on the management of post-stroke dysphagia suggest to include patients in clinical trials considering the low scientific evidence of current data [108].

- **For the prevention of AP recurrences, scopolamine is not recommended (Grade C-1).**

Arguments:

A 2022 Cochrane Library systemic review on the use of scopolamine in hypersalivation of patients with motor neuron disease reported that data were of low scientific evidence to conclude on its benefit effect [109]. Most data come from case series or isolated clinical cases. Scopolamine is also associated with local (irritations) and general adverse events (neurological and psychiatric disorders) [110,111].

- **Drugs needed to be newly assessed and if possible discontinued to avoid AP recurrence are:**
- **Attention-altering drugs (psychotropic drugs) (Grade B-1).**

Arguments:

Conventional or unconventional antipsychotic drugs and benzodiazepines or related hypnotic drugs are associated with an increased risk of pneumonia [27,112–115]. Often chronic, their use is associated with frequent misuses, particularly in the elderly, and should lead to a systematic reassessment of their indication [116].

- **Drugs interfering with swallowing (anticholinergic drugs) ((grade C-1).**

Arguments:

Anticholinergic treatments are associated with swallowing disorders after a stroke and with an increased risk of pneumonia [117–119]. They are also associated with a higher risk of in-hospital death or long-term death following pneumonia [120].

It is thus important to avoid introducing them and to review their indication in patients at risk of AP.

- **Proton-pump inhibitors (Grade B-1).**

Arguments:

Proton-pump inhibitors are often prescribed long-term without any valid indication, especially in the elderly. Such prescription is associated with an increased risk of AP [119,121,122]. When assessing the AP patient's treatment, their indication must be systematically re-assessed so as to discontinue them whenever necessary.

8. Prevention chapter: non-drug therapies

Question 14: Which non-drug measures can be proposed for prevention in patients who experienced AP?

Question 14.1 What postural advice should be offered to patients treated for AP?

Recommendation 14.

Certain postural measures are recommended to prevent AP onset:

- **Positioning the patient at > 30°**, especially patients with enteral nutrition or attention disorders **(Grade B-1)**.
- **Specific movements when swallowing** such as the swallowing maneuver with the chin lowered towards the sternum **(Grade C-1)**.
- **Raising patients to a vertical position and early rehabilitation (Grade B-1).**
- **Mobilization, armchair seating (Grade C-1).**
- **Seating up for meals (Grade C-1).**

Arguments:

The most effective preventive posture is the half-seated position when the patient is bedridden, particularly in patients with enteral nutrition or attention disorders. For nasogastric tube feeding, a slow flow rate (25–50 ml/hours) is suggested at start-up, and gradually increased according to clinical tolerance, up to a maximum of 125 ml/hours. The low cost of preventive postural measures and the absence of any significant harmful effects argue in favor of their implementation. The relevance of this proposition varies from one patient to another. Individualized assessment and treatment should therefore be proposed by speech therapists/phoniatrists or by rehabilitators qualified in the management of swallowing disorders whenever possible for eligible patients [123,124].

Raising patients to a vertical position and rehabilitation as early as possible, in the week following hospital admission, may improve the functional prognosis of patients with AP and reduce the in-hospital mortality rate [22,125,126].

Question 14.2 What measures can be suggested for oral intake (drink, food, etc.) after AP?

Recommendation 14.2 **It is recommended to test swallowing before starting an oral intake (liquid or solid, including medications) (Grade C-1).**

Arguments:

Fasting is associated with a risk of undernourishment, which increases the risk of subsequent swallowing disorders [24,127]. Fluid intake can be compensated for by the venous or subcutaneous route, depending on the patient's needs and venous capital. A rapid assessment of swallowing and nutritional needs must be carried out to limit fasting, especially as fasting does not prevent micro-aspirations, particularly salivary micro-aspirations [22]. Early parenteral nutrition can improve in-hospital mortality over a short period while waiting for enteral feeding to be resumed [128]. However, difficulties with venous access, the risk of hemodynamic overload, electrolyte imbalance, infections, or refeeding syndrome limit its use in frail or elderly patients.

- **It is recommended to adapt the volume and texture of food to the swallowing disorders identified. Thickening liquids can make oral intake safer by reducing aspiration (Grade C-1).**

There is insufficient data on the impact of feeding on oropharyngeal dysphagia in elderly patients to suggest a particular type of feeding, including in patients with dementia [129–131].

It is important to carry out an overall assessment of patients (vital prognosis, nutritional status, cognitive status, impaired judgment, behavioral modification, functional independence) and of the mechanism of AP (attention disorders, chewing disorders, eating disorders or gluttony, swallowing disorders) before suggesting a change in diet.

The diversity of diseases and conditions that can lead to swallowing disorders means that there is no such thing as a standard diet [132]. Thickening liquids to a nectar or honey texture can make oral intake safer by reducing aspiration [131,133–135] (Grade C). Altering the texture of food and beverages, however, runs the risk of reducing fluid and caloric intake (in the case of liquid caloric supplements), as well as impairing the quality of life in the long term.

Various solutions can be proposed, such as splitting meals into smaller bites, modifying textures (soft, crushed, blended), combining foods (gelling, thickening, etc.), and supervising meals [130,136,137]. Temperature changes (cold/hot) or taste enhancers (spices, mint) can help swallowing [24,136,138].

Although classically proposed, there is no scientific evidence to support the benefits of sparkling water in stimulating attention. It is important to obtain the patient's support and those around him/her, and sometimes even to abandon measures that are too restrictive, to ensure correct compliance and avoid the risk of undernutrition due to anorexia. The patient's vital prognosis and quality of life are also important considerations in the context of swallowing disorders [130]. The benefit/

risk ratio must be regularly reconsidered.

Question 14.3: Does oral hygiene prevent new AP onset?

Recommendation 14.3

Oral hygiene (brushing teeth after every meal, daily cleaning of dentures, mouthwash, tongue brushing) tends to reduce the risk of pneumonia, including AP (Grade B-1).

The reduced risk of AP may be explained by the reduction in bacterial inoculum of oral origin in case of salivary aspiration or micro-aspiration [139]. Oral hygiene also helps to improve chewing, bolus (moistening), and swallowing. A recent *meta*-analysis [140] showed the benefit of twice-daily dental care in preventing pneumonia. Considering the absence of harmful effects associated with these body hygiene measures, and despite their low level of scientific evidence, their implementation should be promoted [2,22,108,119,129,140–143]. These measures should be pursued regardless of age to limit dental plaque and periodontal diseases, which are associated with an increased risk of pneumonia.

The systematic use of chlorhexidine mouthwashes should be avoided, due to the increased risk of death, particularly in ventilated patients, and the absence of any obvious additional benefit, particularly in nursing homes, as compared with usual care [2,144]. Dental care also applies to dentures, which must be cleaned with soap and rinsed.

Question 14.4: Does the use of an enteral feeding tube (nasogastric tube, percutaneous endoscopic gastrostomy) help prevent AP?

Recommendation 14.4

At the acute phase, the use of enteral nutrition by nasogastric tube does not prevent AP recurrence (Grade C-1).

In the long term, the use of a feeding tube (whether it be nasogastric tube or percutaneous endoscopic gastrostomy) is a risk factor for aspiration. Feeding tubes ensure adequate nutrition for patients. It is not indicated in patients with advanced cognitive impairment, in end-of-life patients, or in elderly patients with severe malnutrition, as it does not improve survival and impairs quality of life (grade A) [2,119].

Question 15: Which tests should be performed in patients who experienced AP?

Recommendation 15

- **Medical history including the list of treatments and a thorough clinical examination must be performed to look for the main etiologies and risk factors for AP.**
- **A swallowing test and a nutritional assessment are recommended.**
- **Depending on the cases, the following are also recommended:**
  - o **Consultation with a speech specialist/phoniatrist,**
  - o **Consultation with a specialist (geriatrics, ENT, gastro-enterology, neurology),**
  - o **Additional examinations, such as brain imaging, video-fluoroscopy, or video fluoroscopic swallowing examination could be considered.**

Yoshimatsu's checklist can be used to help with the systematic tests to perform for the management of AP (Table S4) [145].

Question 16: Must AP occurrence lead to a change in the patient's treatment commitment?

Recommendation 16

- **It is recommended to discuss the patient's advance decisions with the patient and his or her trusted support person, in order to determine the level of therapeutic commitment in case of AP recurrence in a polypathological patient.**

Arguments:

AP is associated with a poor short-term [146] and long-term [147] prognosis. The 30-day mortality is estimated at 17–31 % depending on the studies, i.e., twice the mortality rate of non-aspiration pneumonia. The 1-year mortality is estimated at 50–65 %, even though this excess mortality is supposedly linked to the significant comorbidities and not to the infectious event in itself [147,148]. Considering the severe prognosis and the high risk of recurrence after a first episode of AP [52], the experts recommend to systematically plan the care pathway asking patients for their advance decisions and the medical team for the level of medical intervention required in case of recurrence [149].

For end-of-life patients, the benefit of the antibiotic therapy for the treatment of a bacterial infection and in particular for a lung infection is questionable. Effectiveness on dyspnea or respiratory congestion control has not been demonstrated. The use of antibiotics at the end of life can lead to adverse events and contribute to antibiotic resistance. Their use should be discussed on an individual basis, with systematic reassessment after 48 h and discontinuation in the absence of symptoms improvement [150–152].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.idnow.2025.105081.

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